

MUHAMMAD NUR AKMAL BIN MOHAMAD RAZIF

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PROFESSIONAL SUMMARY

Full Stack Developer and automation engineer with 2+ years delivering enterprise-grade pipelines at UKM, including a 50,000-record OCR-to-SharePoint migration, SAP real-time integration, and a medallion data architecture in Microsoft Fabric. Experienced across IoT monitoring (Laravel, Django, Node-RED), workflow automation (n8n, Power Automate), and full-stack web development. Seeking a backend, automation, or full-stack role in a product or enterprise environment.

SKILLS

Programming	: Python, HTML, CSS, JavaScript, PHP, PySpark, Dart
Frameworks	: Django, Tailwind CSS, Flask, Laravel, Livewire, Alpine.js, Bootstrap, Flutter
Databases	: PostgreSQL, MongoDB, MySQL, Supabase
Tools & Infrastructure	: Git, Docker, Linux, CI/CD, RESTful API
Automation Tools	: n8n, Node-RED, Power Automate, Microsoft Fabric
Cloud	: AWS

WORK EXPERIENCES

Full Stack AI Developer

May 2025 – Present (Contract)

Fakulti Teknologi dan Sains Maklumat, Universiti Kebangsaan Malaysia, Bangi

- Independently architected enterprise automation workflows using n8n, Power Automate, and Microsoft Fabric, designing a full Medallion (Bronze → Silver → Gold) data pipeline architecture to transform raw maintenance data into production-ready datasets for downstream analytics and reporting.
- Engineered an automated SAP-to-SharePoint integration via Power Automate, achieving near real-time data synchronization with 5-minute interval updates and eliminating manual cross-system data entry that previously required significant manual effort.
- Supervised a part-time validation team of 30+ members, establishing structured review workflows to catch and correct OCR extraction errors before ingestion into SharePoint, ensuring data quality across 50,000+ migrated records.
- Developed the backend for a data extraction and validation system, supporting an OCR-driven pipeline that migrated 3.5 years of historical maintenance records into structured SharePoint, enabling predictive analytics for the first time.

Full Stack Developer

October 2024 – March 2025

Crescita Technology (M) Sdn Bhd, Cyberjaya

- Developed a cross-platform IoT monitoring ecosystem integrating 6+ sensor types (temperature, humidity, CO₂, and others) with a centralized Django web dashboard, providing real-time operational visibility across a proof-of-concept multi-site deployment.
- Built a cross-platform mobile monitoring application in Flutter/Dart, consuming RESTful APIs to stream live sensor readings and display real-time status across multiple device types.
- Designed Node-RED pipelines to acquire, buffer, and store time-series sensor data in MongoDB, enabling historical replay and trend analysis, replacing a previously manual logging process.

Research Assistant (Computer Vision) Intern

September 2023 – January 2024

Aerospace Malaysia Innovation Centre (AMIC), Kajang

- Built and integrated an OS2D-based aircraft part detection model, improving identification accuracy from 50% to 65% and reducing detection time to under 10 seconds.
- Developed automated spray path optimization for the UR10 robot by coupling detection output directly to the pathfinding algorithm.

PROJECTS

Condition of Switch Monitoring System (COSMOS) | Laravel, Livewire, Tailwind CSS, Alpine.js, PostgreSQL, MongoDB

- Built a real-time monitoring dashboard to visualize track switch sensor data across 3 MRT stations, enabling centralized operational visibility and faster fault detection for maintenance teams.
- Developed custom canvas-drawn threshold band annotations using uPlot for high-performance time-series rendering, supporting toggle-visibility legends and CSV export.

Maintenance Checklist OCR Extraction and Validation System | Laravel, Livewire, Tailwind CSS, Alpine.js, PostgreSQL, n8n

- Architected end-to-end automation workflows using n8n, migrating 50,000+ historical maintenance records across 4 engineering departments into structured SharePoint, enabling predictive analytics for the first time.
- Developed a high-performance data entry web application using the TALL stack, integrating with the automation pipeline above, and oversaw a validation team of 30+ that revalidated the extracted data.

Plant Pollen Detection Dashboard | Python, Flask, HTML, CSS, JavaScript, Bootstrap

- Engineered an automated pollen detection dashboard integrating a pre-trained ML model to process batch image folders and render bounding box visualizations.
- Implemented export of viable/non-viable pollen metrics to Excel, reducing analysis time from hours to minutes for research staff.

Indeks Tadbir Urus Koperasi | Flutter, Dart, Supabase

- Developed a cross-platform mobile application enabling cooperative organizations to conduct structured governance and management surveys with dynamic questionnaires and a performance scoring system.
- Built chart-based visual analytics screens to display survey results, allowing administrators to benchmark cooperative governance performance over time.
- Implemented PDF report export from survey results, enabling efficient sharing and documentation for regulatory and audit purposes.

Mushroom Grow House Monitoring and Sales Management System | *Python, Django, PostgreSQL, MongoDB, Node-RED*

- Developed a real-time environmental monitoring platform for mushroom grow houses, tracking live sensor data across temperature, humidity, and CO₂ parameters through a unified dashboard.
- Implemented a sales and inventory module with automated PDF generation for invoices, receipts, and shipping documents, streamlining order fulfilment and reducing admin workload for grow house operators.

Reservoir Monitoring and Management System | *Python, Django, PostgreSQL, MongoDB, Node-RED*

- Built a real-time web-based reservoir monitoring platform using Django and Python, providing live sensor data visualization for early issue detection, maintenance prioritization, and data-driven decision-making.
- Integrated Node-RED for automated sensor data acquisition and valve control based on real-time conditions, deployed and maintained on IIS (Windows Server 2022) for high availability and scalable production performance.

Getahpedia : Rubber Leaf Disease AI Detection | *Flutter, Dart, Python, TensorFlow*

- Built a cross-platform mobile application integrated with a fine-tuned ResNet50 deep learning model for accurate rubber leaf disease detection and classification.
- Implemented an encyclopaedic knowledge base covering rubber tree leaf disease, empowering users to identify, learn about, and reference disease types independently.

EDUCATION

Bachelor of Computer Science with Honors (CGPA: 3.78/4.00) Universiti Kebangsaan Malaysia (UKM)	October 2020 – August 2024
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CERTIFICATIONS

- AWS Certified Cloud Practitioner (Amazon Web Services Training and Certification, 2025)
- Google Advanced Data Analytics, (Coursera, 2024)

REFERENCE

References available upon request